

Simulation and reconstruction chain of the inclusive and coherent $\cos 2\phi$ measurements with a tensor-polarized ^6Li beam at the EIC: assumptions, methodology and closure

2026-08-24 · rewritten 2026-08-28 after a full review of the toolchain (Appendix A) · companion to Report 1 (the projections) and Report 3 (the machine and the detector) · computed with `polligen.reco`, `polligen.hfs` and `polligen.recopseudo` on the PYTHIA 8 hadronic final state of `tools/pythia8`; every number is reproducible with the commands of `docs/reproduction_manual.md`

Abstract. The double-helicity-flip structure function $\Delta(x, Q^2)$ of ^6Li is projected in Report 1 as the $\cos 2\phi$ modulation of inclusive DIS on a transversely tensor-polarized beam, and as a modulation of the recoil azimuth in coherent diffraction with the ^6Li detected intact. This report describes the simulation and reconstruction chain behind those projections — the polarized event kernel, the PYTHIA 8 hadronic final state, the electron-side, hadron-side and far-forward detector responses, the kinematic reconstruction, the spin-state-sorted estimator and the conversion to Δ — lists every assumption it rests on with its provenance and leverage, and reports its closure at the reconstructed level. The analysis path consumes only quantities an experiment records — the spin state of each bunch crossing, the tensor polarization and luminosity of each state, the reconstructed electron, the hadronic $E - p_z$ sum and the Roman-Pot angle pair — with simulation-derived corrections applied as functions of reconstructed variables; a 2026-08-28 review of the toolchain found one place where that was not so (the hadronic-scale calibration, now keyed on the reconstructed point), a truncation of the PYTHIA library below $x = 16/s$ (now regenerated), and two statistical defects of the estimator (now fixed), none of which changes a headline number by more than its statistical error. At $e(10 \text{ GeV}) \times ^6\text{Li}(99.5 \text{ GeV/u})$ the four sweet-spot bins reconstruct with purities 0.56–0.75 and amplitude errors $0.9\text{--}3.0 \times 10^{-4}$ in one year, the fit closing on the response-folded truth; the coherent channel closes at the lithium tagging optics of Report 1 with $2.5\text{--}6.0 \times 10^6$ tagged recoils per year, unbiased in every $|t|$ bin with more than 10^5 recoils.

1 · Scope

Report 1 presents the measurement; Report 3 the accelerator and detector it is projected on. This report is the layer between them: what the simulation generates, what it pretends the detector records, what the analysis is allowed to read, and how well the chain returns what was put in. Section 2 walks the chain stage by stage with the boundary between generator truth and measured quantities made explicit; Section 3 collects the assumptions; Section 4 gives the methodology of each step; Section 5 reports the reconstructed-level closure; Section 6 the systematic effects that were measured on the chain and the specifications they imply; Section 7 what the chain does not model. The audit of the code that produced this revision is summarized in Appendix A and recorded in `docs/code_review_2026-08-28.md`.

2 · The chain

The observable is a Fourier coefficient of a *ratio* of spin-sorted azimuthal distributions. For an unpolarized electron on a spin-1 target with alignment axis at polar angle θ_S and azimuth ϕ_S about the virtual photon, the Hoodbhoy–Jaffe–Manohar cross section of the projection state m is [1,2]

$$\frac{d\sigma^{(m)}}{dx dy d\phi} = \frac{2y\alpha^2}{Q^2} \left[F_1 + \frac{2}{3}a_m b_1 + \frac{1-y}{xy^2} \left(F_2 + \frac{2}{3}a_m b_2 \right) - \frac{1-y}{y^2} c_m \sin^2 \theta_S \Delta(x, Q^2) \cos 2\phi \right], \quad c_m = 3m^2 - 2,$$

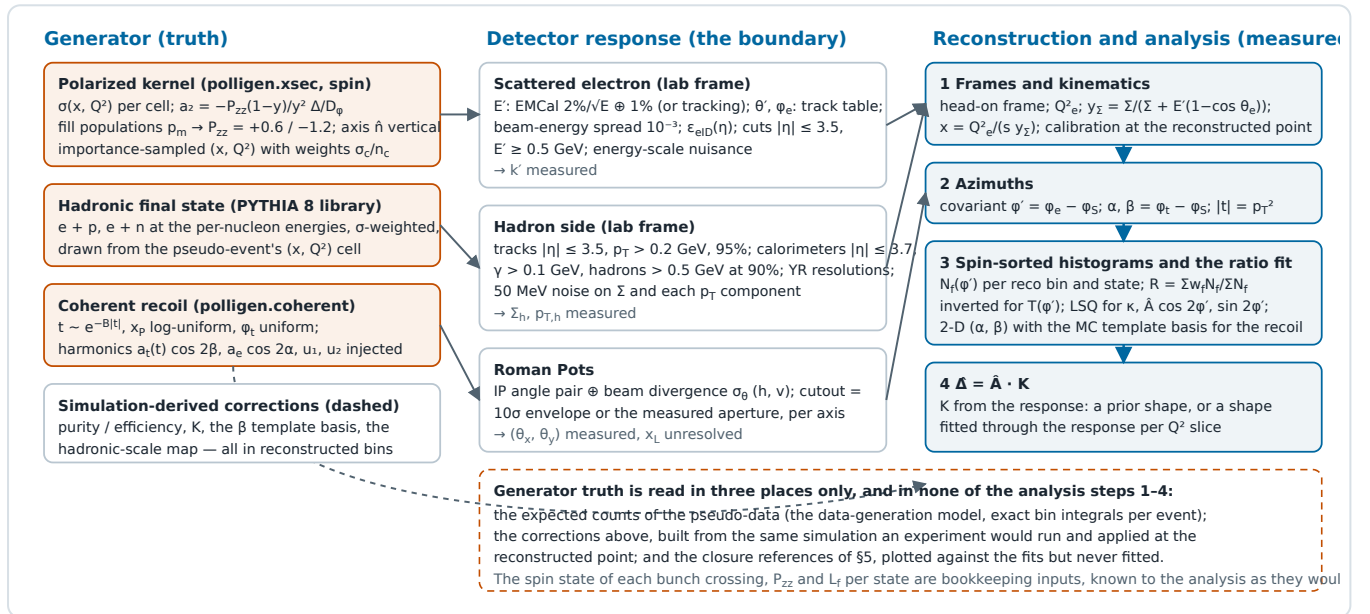
so that a fill with populations p_m along a transverse axis has the azimuthal distribution $1 + a_2 \cos 2\phi'$, $\phi' = \phi - \phi_S$, with

$$a_2 = -P_{zz} \frac{1-y}{y^2} \frac{\Delta}{D_\phi}, \quad D_\phi = F_1 + \frac{1-y}{xy^2} F_2, \quad \hat{A} \equiv \frac{a_2}{P_{zz}}, \quad P_{zz} = \sum_m p_m c_m.$$

The chain that turns this into a measurement has six stages (Schematic 1, Table 1). **(i) The polarized kernel.**

`polligen.xsec` evaluates the master formula per (x, Q^2) cell for each projection state m of the fill, with the spin-density

matrix of `polligen.spin` providing the m-populations of the two fill types and the covariant azimuth of §4.1 providing ϕ' ; the unpolarized cross section, F_2 and $R = \sigma_L/\sigma_T$ come from `polli_fastsim.structure` and the Δ model from `polli_fastsim.delta_models`. **(ii) The hadronic final state.** The tensor polarization enters only the azimuthal weight of the inclusive cross section, so the hadronic system of each pseudo-event is taken from an unpolarized PYTHIA 8 library of $e + p$ and $e + n$ events at the same beam energies (§4.2), weighted by their cross sections. **(iii) The detector response.** The scattered electron is smeared in the detector frame with the calorimeter, tracking and electron-identification models of §3; the hadronic final state passes through a hadron-side response — tracker and calorimeter coverage, thresholds, efficiencies, Yellow-Report resolutions, calorimeter noise — in the same frame; the coherent recoil is measured by an emulated Roman-Pot system that returns the angle pair at the interaction point smeared by the beam divergence and cut by the near-beam envelope. **(iv) Reconstruction.** The head-on frame is restored, Q^2 comes from the electron, y from the Σ method on the hadronic $E - p_z$ sum, x from the two (the mixed method), and the azimuths from the covariant formula on the reconstructed four-vectors. **(v) The estimator.** Events are histogrammed in ϕ' per spin state and per reconstructed (x , Q^2) bin, and the $\cos 2\phi'$ coefficient is fitted to the spin-state ratio, which cancels any azimuthal acceptance common to the two states. **(vi) Extraction.** The amplitude is converted to Δ with a bin-centering factor derived from the response — either from an assumed Δ shape, or from a shape fitted through the response itself.



Schematic 1 — the chain, and the boundary between truth and measurement. Left: what is generated. Middle: the three detector responses, which consume the truth and emit measured quantities. Right: the analysis, which reads measured quantities, the bookkeeping of the run plan, and corrections derived from the simulation in reconstructed bins.

STAGE	INCLUSIVE $E' \text{ LI} \rightarrow E' X$	COHERENT $E' \text{ LI} \rightarrow E' X \text{ LI(G.S.)}$	TRUTH READ	CODE
Generation	importance-sampled (x, Q^2) over a 60×45 log grid with the generator window loosened ($y \geq 0.004$, $Q^2 \geq 0.7$ GeV ² , $ \eta \leq 3.8$, $E' \geq 0.3$ GeV) so events can migrate into the analysis window; 400 pseudo-events per cell, weight σ_c/n_c	$ t $ exponential with $B = 50$ GeV ⁻² , $x_p \in [10^{-3}, 10^{-2}]$, ϕ_t uniform; 6×10^6 recoils; injected harmonics of §4.5	—	recopseudo.RecoResponse, CoherentResponse
Hadronic final state	PYTHIA 8.311 $e+p / e+n$ at the per-nucleon energies, $Q^2 \geq 0.7$ GeV ² , $\hat{m} \geq 0.5$ GeV, lepton radiation off; 8 M events over the three configurations; a library	not generated (no electron side; α integrated analytically)	the pseudo-event's (x, Q^2, y) , to draw and scale the library event	hfs.HFSLibrary, HFSResponse; tools/pythia8

	event from the pseudo-event's (x , Q^2) cell supplies the captured fraction of Σ_h and the p_T ratio			
Detector response	e^- : lab-frame smearing of E' , θ' , ϕ_e , beam-energy spread, $\epsilon_{elD}(\eta)$; hadrons: coverage, thresholds, efficiencies, resolutions, noise, in the lab frame	Roman Pots: angle pair $\oplus$ divergence, rectangular cutout (envelope or aperture per axis)	the true four-vectors, to smear them	<code>reco.smear_electron</code> , <code>hfs.HadronResponse</code> , <code>reco.rp_measure</code>
Reconstruction	head-on frame; Q^2_e ; y_Σ (JB, DA available); $x = Q^2_e/(s y_\Sigma)$; hadronic-scale calibration at (x_{mixed} , Q^2_e); analysis cuts $Q^2 \geq 1$, $0.01 \leq y \leq 0.95$, $W^2 \geq 10$, $ \eta \leq 3.5$, $E' \geq 0.5$; covariant ϕ'	$ t = p_T^2$, $\beta = \phi_t - \phi_S$ from the smeared angle pair; $\alpha = \phi_e - \phi_S$	none	<code>reco.electron_method</code> , <code>hfs.hadronic_kinematics</code> , <code>reco.azimuth_wrt_lepton_plane</code>
Pseudo-data	exact expected counts per spin state and ϕ' (or (α , β)) bin, from the per-event amplitude at the true kinematics and the reconstructed bin assignment, with the ϕ' -efficiency harmonic (common or per fill) and the relative-luminosity offset switched on; Poisson-fluctuated at 10 and 100 fb ⁻¹ /u		the per-event a_2 and dilution (data generation)	<code>RecoResponse.expected_counts</code> , <code>CoherentResponse.expected_counts_2d</code>
Estimator	spin-state ratio per bin, inverted for $T(\phi')$, weighted LSQ for κ , $\hat{A}$, $\hat{A}_{sin}$	the same in (α , β) with the acceptance-weighted β template basis from the response, the u_1 , u_2 nuisances subtracted, and three parity-forbidden sine columns as null tests	the basis and the truth references only	<code>reco.harmonic_ratio_fit</code> , <code>harmonic_ratio_fit_2d</code>
Extraction	$\Delta = \hat{A} K$, $K = \Delta_{prior}(x_c, Q^2_c)/\text{fold}(\Delta_{prior})$ or from the shape fitted through the response	$a_t(t_{ref})$, a_e per $ t $ bin	the response (fold, purity, efficiency)	<code>recopseudo.fold</code> , <code>fold_shape_fit</code> , <code>delta_from_amplitude</code>

Table 1 — the chain stage by stage, and where generator truth is read. The pseudo-data are generated from the truth, as data are generated by nature; the reconstruction and the estimator read measured quantities; the corrections are what an experiment takes from its own simulation.

3 • Assumptions

Table 2 lists every input the chain consumes, with its value, its provenance and what a wrong value would move. Three classes are distinguished: inputs the accelerator or the detector will fix (to be asked for precisely; Report 3 carries their full specification), physics-model inputs (scenarios, bracketed by bands in Report 1), and idealisations of this analysis (to be replaced or quantified as the chain matures). Idealisations that are structural — no radiative corrections, no backgrounds, no coherent electron side — are in §7.

INPUT	VALUE USED	PROVENANCE AND STATUS	LEVERAGE IF WRONG
Beam and spin			

${}^6\text{Li}$ energies	40.8 / 99.5 / 137.5 GeV/u	derived: γ -matched to the 41 / 100 / 275 GeV proton, rigidity-capped at the top [7,13]; the chain publishes at the mid configuration	every far-forward cut ($\propto A p_u$); the sweet-spot y values
beam divergence σ_θ (h / v)	220/380, 180/180, 92/92 μrad	YR Table 10.1 proton values, gamma-matched species step [7] (reco.sigma_theta_for); no lithium optics is published	coherent tag acceptance, exponentially; $\Delta p/p$ 7-10 $\times 10^{-4}$ enters only the pot dispersion term
tagging optics (coherent channel)	$\beta_x^* \times 50$ / 176 / 89, pots following the 10 σ envelope 0.33 $\times$ 3.8, 0.17 $\times$ 1.8, 0.12 $\times$ 0.92 mrad	this programme's proposal (Report 1 §6.1): the optimum of acceptance $\times$ luminosity with the vertical plane at high acceptance, $L/L_{\text{HA}} = 1/7.1, 1/13.3, 1/9.5$; assumes parallel-to-point transport and pots that move	without it no recoil is tagged at any configuration (Figure 2c)
P_{zz} of the two spin states	+0.6 / -1.2, equal luminosity	in-ring placeholder (ANL source target ≥ 0.8); the $m = 0$ -rich state at $-2P_{zz}$ assumes the same source purity; taken as exactly known by the analysis	statistical error $\propto 1/P_{zz}$; a scale error is a 1:1 normalisation on Δ (§6)
spin axis	vertical, $\varphi_S = \pi/2$, $\theta_S = 90^\circ$	the stable spin direction of the arcs; an azimuthal error is self-calibrated by the $\sin 2\varphi'$ term (tan 2 δ recovered exactly in a test), a polar tilt dilutes by $\sin^2\theta_S$	0.8% dilution at 5° tilt; none from the azimuth
spin-state pattern	bunch-by-bunch alternation	required, not assumed: the ratio cancels only an acceptance common to both states (§4.3)	a 10^{-3} difference of the $\cos 2\varphi'$ acceptance harmonic between the states fakes 5.6×10^{-4}
luminosity	10 / 100 fb^{-1} per nucleon	YR convention for one / ten years [7]; shares 0.5/0.5 with a 10^{-3} offset unknown to the analysis	statistics; the offset enters the ratio as a constant

Central detector

EMCal σ_E/E for e'	2%/ $\sqrt{E}$ @ 1%	YR requirement of the backward crystals [7]; correct where the sweet-spot electrons sit ($\eta = -2.9$ to -1.6); the η -dependent table changes Δ by $\leq 0.1\%$	$\delta Q^2/Q^2$ (subdominant to the angles)
tracking σ_p/p	$\sqrt{(a p)^2 + b^2}$, η -piecewise	YR requirement (Fig. 8.3 / Table 11.2), endcap constants padded; a requirement, not an achievement	only with energy="best"
track angular resolution	5 / 3 / 2 / 1 mrad by η (3 mrad at the sweet spots)	placeholder — no YR angular requirement exists; it is also the only stand-in for the 81-211 μrad beam-divergence floor on θ'	$\delta Q^2/Q^2 = 5.2\%$ at spots 1-2; a factor ≤ 3 bracket (purity 0.66 $\rightarrow$ 0.70 at a third of it)
electron identification	$\epsilon_{\text{eID}}(\eta) = 0.70\text{--}0.95$	constructed between the ATHENA and ECCE proposal values; no ePIC curve; no purity or background model	none on $\hat{A}$ beyond statistics — a fill-independent efficiency cancels in the ratio (an η tilt of 0.05 moves Δ by 0.02%)
electron energy scale	nuisance $\pm 1\%$	unknown to the analysis; $d \ln x/d \ln E' = 2 - y$ for the Σ method	0.3-1.0% on Δ per 1%
hadron-side coverage and thresholds	tracks $ \eta \leq 3.5$, $p_T > 0.2$ GeV, 95%; calorimeters $ \eta \leq 3.7$; $\gamma > 0.1$ GeV; hadrons > 0.5 GeV, 90%	stand-ins of the right magnitude in the detector frame (the ePIC nominal forward reach is 4.0; B0 at $\eta = 4.6\text{--}5.9$ not modelled)	the captured fraction of Σ_h (0.70-0.85 at the sweet spots, Table 3); 4.0 recovers a quarter of the forward loss with $\delta y/y$ unchanged
hadron-side resolutions	YR tables per region	requirement values [7], with the YR 2% EMCal constant	subdominant to the noise below $y = 0.02$
calorimeter noise on Σ_h	50 MeV Gaussian per event on Σ and each p_T component	a one-parameter stand-in for the noise and threshold floor whose leverage at low y Arratia	the $y \approx 0.01$ resolution: 0.22 $\rightarrow$ 0.54 $\rightarrow$ 1.01 at $y = 0.005$ for $0 \rightarrow 50 \rightarrow 100$ MeV

et al. show for H1/ATHENA [20]; no ePIC number exists (plans/04 #21)			
hadronic energy scale	calibrated from the simulation at (x_{mixed}, Q^2_e); residual $\pm 2\%$ as a nuisance	the analysis's own calibration; keyed on reconstructed variables (2026-08-28)	0.1–0.6% on $\hat{A}$ per 1% residual
Far forward			
Roman-Pot measurement	angle pair $\theta \ \sigma_\theta$ (h, v); no detector term	beam effects dominate the ePIC far-forward resolution [17]; x_L unresolved; the dispersion of the off-momentum recoil ($x_p \leq 0.01$, below every divergence) is folded into the horizontal term of the tagging optics	$ t $ smearing $2p_T \delta p_T$; the ϕ_t dilution the template basis absorbs
pot aperture	2.00/3.00, 1.35/3.00, 1.03/2.30 mrad (h/v)	measured with an intact ^6Li in a September-2024 ePIC geometry (<code>tools/fullsim</code>); the pots have moved since	with fixed pots the tag is 10^{-8} – 10^{-26} ; the tagging optics assumes pots that follow
exclusivity	pure coherent sample	no breakup background, vetoes or Z-identification modelled (plans/06); the incoherent yield is m-state blind and dilutes rather than fakes	the $ t $ -fit purity of Report 1 §6.4
Physics model			
F_2, F_1, R	toy F_2 ($\pm 37\%$ of CT18 over the acceptance); $R = 0.18/(1 + Q^2/50)$	placeholders with PDF-grid and R1998 backends behind the same hooks (Report 1 §3.2); F_2 cancels exactly in the amplitude, R does not	rates and statistical errors to $\times 1.5$; Δ/F_1 by $+18 / +19 / +8 / -4\%$ at the four spots if R1998 replaces the placeholder
$\Delta(x, Q^2)$	moment-constrained ansatz A, dilution 1/3; α_s from the CT18NLO table	Report 1 §4; the LO α_s fallback (used without the <code>parton</code> package, now announced) raises the normalisation by 14%	the injected amplitudes, not the closure
coherent scenario	$f_0 = 0.04$, $B = 50 \text{ GeV}^{-2}$, $\varepsilon_{B0} = -0.08$, $a_e = 0.01$, $u_1 = 0.05$, $u_2 = 0.02$	Report 1 §6; u_1, u_2 at the ZEUS 1σ edge [13,14], taken as known by the fit	rates and the injected harmonics; δu_2 leaks into a_e at first order (§6)
$a_t(t)$ template	linear in $ t $	the anchor's low- $ t $ behaviour [9]; one coefficient per $ t $ bin	a different shape is a model systematic to be varied

Table 2 — the assumptions. "Requirement" means the Yellow Report value, which is a design target and not a measured performance. Every number in this report is conditional on this table.

4 • Methodology

4.1 • Frames and azimuths

The generator works in the head-on frame (ion along $+z$, electron along $-z$); the detector sees the lab frame, in which the electron beam defines the ePIC axis and the ion enters at -25 mrad in x . The two are related by a rotation of 12.5 mrad about y and a boost $\beta_x = \sin 12.5$ mrad; energies change by 8×10^{-5} and the vertical spin direction not at all. Every detector effect — the electron smearing, the hadron acceptance and thresholds, the Roman-Pot angles — is applied in the lab frame, and the measured four-vectors are transformed back before any kinematic quantity is formed, as an ePIC analysis does. For the electron the transformation changes only odd harmonics of the azimuth ($\approx 10^{-3}$ in $\cos \phi$, $\cos 2\phi$ untouched to 10^{-17}), so it cannot fake the observable; for the hadrons it moves the forward calorimeter edge by up to ± 0.25 units of η around the ring, which is worth $+0.7$ to $+1.5\%$ of the captured Σ_h at the sweet spots relative to a head-on-frame acceptance.

The azimuth of the observable is the covariant one used for every transverse-spin azimuth in SIDIS and exclusive analyses [3,4], applied to the spin four-vector S and, for the recoil, to P' :

$$\cos \phi_S = -\frac{l_\perp \cdot S_\perp}{l_\perp \|S_\perp|}, \quad \sin \phi_S = -\frac{\epsilon_{\mu\nu\rho\sigma} l^\mu S^\nu P^\rho q^\sigma}{(P \cdot q) \sqrt{1 + \gamma^2} l_\perp \|S_\perp|}, \quad v_\perp^\mu = g_\perp^{\mu\nu} v_\nu, \quad \gamma^2 = \frac{M^2 Q^2}{(P \cdot q)^2}.$$

With the axis transverse at lab azimuth ϕ_S this reduces exactly to $\phi' = \phi_e - \phi_S$ for a massless target and to $O(\gamma^2)$ for the nucleus — below 5 mrad at the sweet spots of the mid and top configurations ($\gamma^2 \leq 0.02$) and 12 mrad at those of the low configuration — and it is verified against an explicit collinear-frame construction to 10^{-13} . $\cos 2\phi'$ is even under $\phi' \rightarrow -\phi'$ and $\phi' \rightarrow \phi' + \pi$ and the axis is headless, so no orientation convention enters; a misalignment δ of the assumed axis turns the modulation into $\cos 2\delta \cos 2\phi' + \sin 2\delta \sin 2\phi'$, and the fitted $\sin 2\phi'$ term returns $\tan 2\delta$ — the self-calibration of the axis. The tensor structure functions b_1 – b_4 leak into $\cos 2\phi'$ at $O(\gamma^2)$ with a bound of $1.15 \gamma^2 b_1/F_1$ (the full Cosyn et al. combination [12]), $\leq 0.15\%$ of everything reported here and subtractable with the A_{zz} of the same data.

4.2 • Kinematic reconstruction and the hadronic final state

The electron method gives Q^2 well ($\delta Q^2/Q^2 = \delta E'/E' \oplus \cot(\theta'/2) \delta\theta'$, 5.2% at the two $Q^2 = 1.14 \text{ GeV}^2$ spots with the 3 mrad angular placeholder, 2.9% and 1.6% at the other two) but loses y as $(1 - y)/y$: three of the four sweet spots sit at $y = 0.010$ – 0.025 (Table 3), where a 1.2% energy resolution gives $\delta y/y = 0.46$ – 1.2 and the bins are not reconstructible. The remedy is HERA's [5,6]: y from the hadronic final state,

$$y_\Sigma = \frac{\Sigma}{\Sigma + E'(1 - \cos \theta_e)}, \quad \Sigma = \sum_{h \neq e'} (E_h - p_{z,h}), \quad x = \frac{Q_e^2}{s y_\Sigma},$$

with θ_e from the ion direction. Σ is insensitive to particles lost along the ion beam (each contributes $E - p_z = m_T e^{-y}$) and y_Σ to collinear initial-state radiation; the mixed x inherits both, because the factor $(1 - z)$ of a radiated photon cancels between Q_e^2 and s — it is the Q^2 label of the bin that migrates under radiation, not x (plans/08 D3 corrected). The Jacquet–Blondel and double-angle methods are implemented alongside and checked against hand-derived formulas; the Σ method is the best of the three below $y = 0.05$ and is used throughout.

The hadronic final state is a PYTHIA 8.311 production of 8 M unpolarized $e + p$ and $e + n$ events at the three per-nucleon beam energies ($Q^2 \geq 0.7 \text{ GeV}^2$, lepton radiation off, $\hat{m} \geq 0.5 \text{ GeV}$ so that the library covers the whole x range of the generator window — its 4 GeV default had silently removed $x < 16/s$, 39% of the selected rate, before 2026-08-28), merged in the ratio of their cross sections. It enters as a library: for a pseudo-event at (x, Q^2) a library event of the same cell, passed through the hadron-side response with its noise off, supplies the captured fraction of Σ_h and the p_T ratio, which are applied to the pseudo-event's true $\Sigma_h = 2E_e y + m_N^2/(E_N + p_N)$ — the massless value plus the same target-mass term the library's Σ carries, 4.4 MeV at 99.5 GeV/u — and the 50 MeV noise is added per pseudo-event; y_Σ then follows with the pseudo-event's own reconstructed electron. Table 3 and Figures 3–4 give the result at the mid configuration: 80–92% of the true Σ_h is within the acceptance and above threshold, 7–17% escapes forward beyond $|\eta| = 3.7$ (the target-fragmentation side of a $W = 6$ – 10 GeV system; the lithium fragments at $\eta \approx 8$ never enter, each carrying only $m^2/2p$), 1–6% lies below threshold, and 70–85% is captured through the full response. The loss is a scale bias on y_Σ , calibrated as an analysis would calibrate it: the library's own events, smeared and reconstructed like the pseudo-events, give a map of $\langle \Sigma_{\text{reco}} \rangle / \langle \Sigma_{\text{true}} \rangle$ in bins of the reconstructed $(x_{\text{mixed}}, Q_e^2)$, and each pseudo-event's sums are divided by the map at its own uncalibrated reconstructed point. (Until 2026-08-28 the map was looked up at the event's *true* cell, which no experiment can do; the measurable version gives purities within 0.06 of it and is the only one used now.) What sets the resolution is not the loss but the noise: with 50 MeV the Σ method gives $\delta y/y = 0.32 / 0.22 / 0.29 / 0.15$ at the four spots against 0.19–0.22 without noise, and at $y = 0.005$ it runs $0.22 \rightarrow 0.33 \rightarrow 0.54 \rightarrow 1.01$ for $0 \rightarrow 25 \rightarrow 50 \rightarrow 100 \text{ MeV}$. Each configuration selects its own four bins; at the low and top configurations they resolve to $0.39 / 0.23 / 0.24 / 0.11$ and $0.23 / 0.19 / 0.21 / 0.18$, so the $(0.141, 14.3 \text{ GeV}^2)$ bin is best measured at 5×40.8 and the low- x bins improve with $\sqrt{s}$.

SWEET SPOT (x, Q^2)	Y	W [GeV]	E': E', Θ', H	ΔY/Y E' ALONE	Γ _H AS H · Σ- WEIGHTED PARTICLE H	WITHIN ACCEPTANCE, H ≤ 3.7	LOST FORWARD	BELOW THRESHOLD	CAPTURED THROUGH THE RESPONSE	ΔY/Y: Σ, JB, DA
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					(16 / 50 / 84%)	(TRACKS / Γ / HCAL)						(MEAN · MEDIAN)
0.028, 1.14	0.010 6.3	9.98 GeV, 107 mrad, -2.93	1.18	1.6 · 1.8 / 2.6 / 3.8	0.80 (0.45 / 0.25 / 0.10)	0.17	0.03	0.70 · 0.73	0.32, 0.32, 0.51			
0.011, 1.14	0.025 10.0	9.90 GeV, 107 mrad, -2.92	0.46	0.7 · 0.8 / 1.6 / 2.9	0.87 (0.51 / 0.26 / 0.10)	0.08	0.06	0.74 · 0.78	0.22, 0.22, 0.40			
0.071, 3.13	0.011 6.5	10.02 GeV, 177 mrad, -2.42	1.06	2.1 · 1.9 / 2.6 / 3.7	0.83 (0.47 / 0.25 / 0.10)	0.15	0.02	0.74 · 0.77	0.29, 0.29, 0.44			
0.141, 14.3	0.025 9.4	10.10 GeV, 378 mrad, -1.65	0.45	2.0 · 1.7 / 2.2 / 2.9	0.92 (0.55 / 0.27 / 0.09)	0.07	0.01	0.85 · 0.87	0.14, 0.15, 0.21			

Table 3 — the four sweet spots of Report 1 at $e(10) \times {}^6\text{Li}(99.5)$ and their hadronic final state (PYTHIA 8, 4 M e + p and e + n events; `hfs_acceptance.py`, `hfs_resolution.py`). The acceptance columns are geometry and thresholds only, in the detector frame; "captured through the response" includes efficiencies, the track turn-on and the mass hypotheses, noise off, and equals the median y_{Σ}/y ; the resolutions are 68% half-widths through the full response with 50 MeV of noise. Differences of 0.01 between this table and the figures are Monte-Carlo noise.

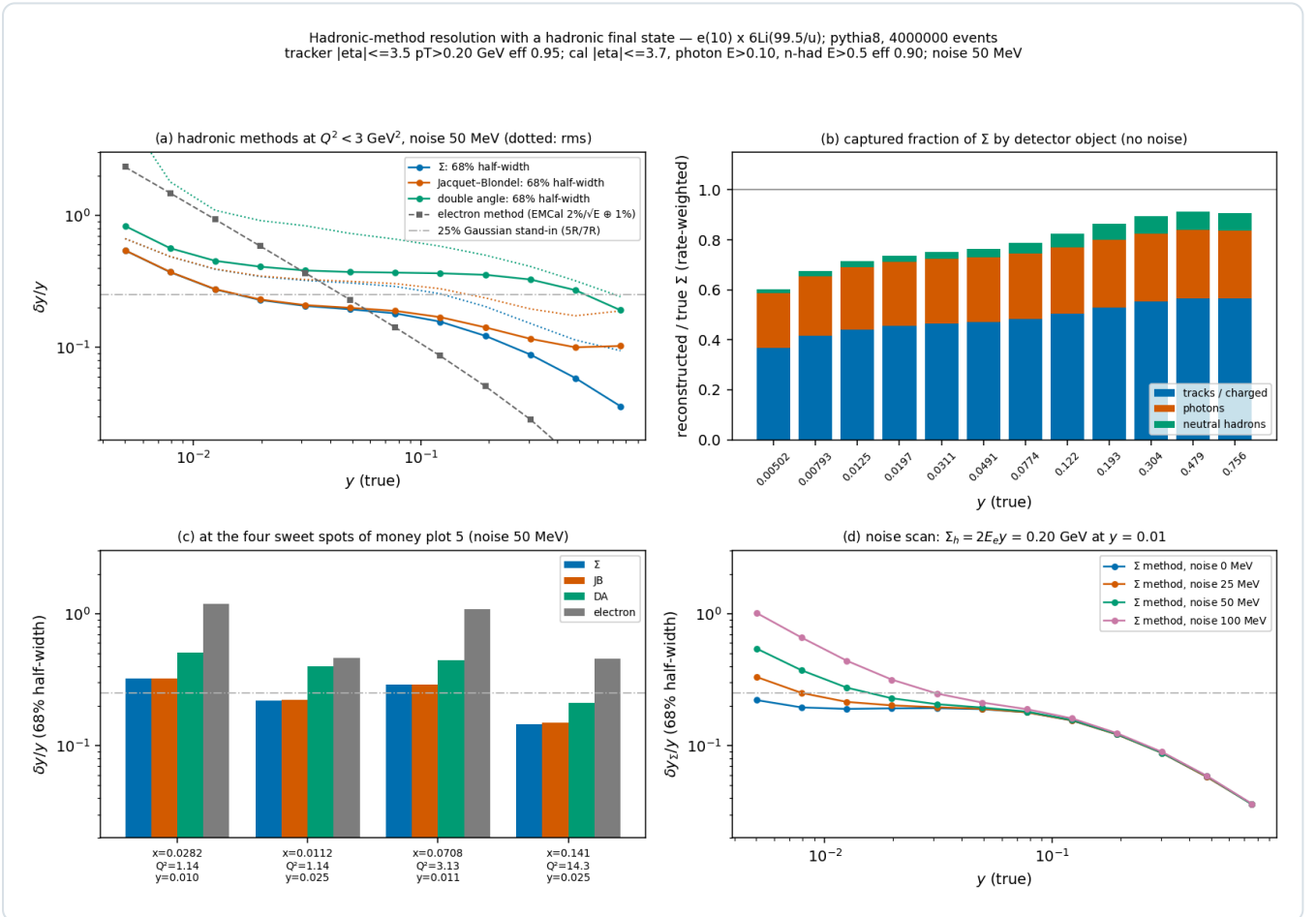


Figure 3 — hadronic-method resolution ($e(10) \times {}^6\text{Li}99.5$, PYTHIA 8, `hfs_resolution.py`). **(a)** $\delta y/y$ (68% half-width; dotted: rms) of the Σ , Jacquet-Blondel and double-angle methods at $Q^2 < 3 \text{ GeV}^2$ with 50 MeV of noise, against the electron method and the 25% stand-in of the earlier chain. **(b)** Captured fraction of Σ by detector object, rate-weighted, no noise. **(c)** The four sweet spots. **(d)** The Σ -method resolution for 0, 25, 50 and 100 MeV of noise: the floor decides the $y \approx 0.01$ bins.

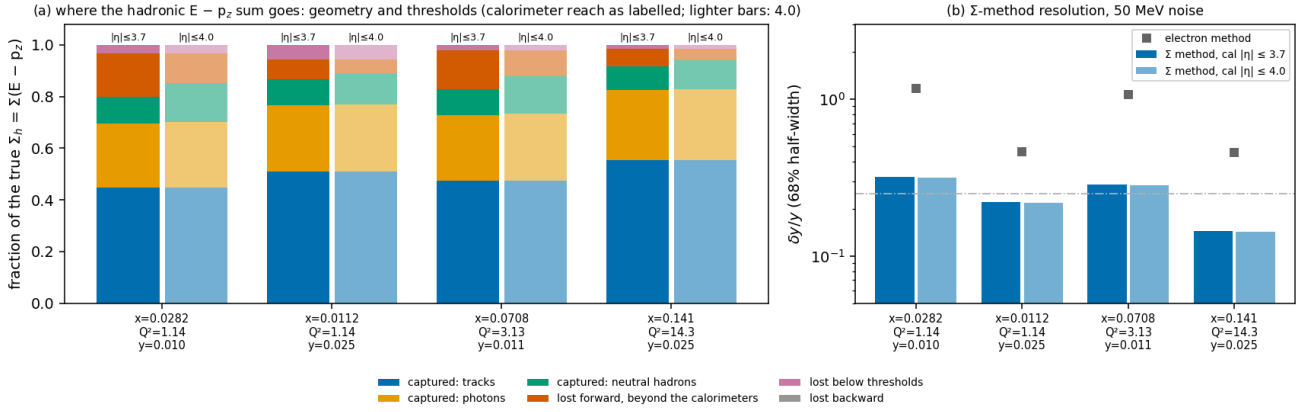


Figure 4 — where the hadronic $E - p_z$ sum goes (e10 x ^6Li 99.5, PYTHIA 8, `hfs_acceptance.py`). **(a)** The true Σ_h by fate — captured as tracks, photons and HCal objects; lost forward beyond the calorimeters, below threshold, or backward — for a calorimeter reach of $|\eta| \leq 3.7$ (full) and the ePIC nominal 4.0 (light). **(b)** The Σ -method $\delta y/y$ through the full response for both reaches: the extra reach moves the bias, not the resolution.

4.3 · The spin-state-sorted estimator

A single-fill fit of $N(\phi')$ measures the detector's own $\cos 2\phi'$ acceptance harmonic divided by P_{zz} on top of the physics: a 3% harmonic aligned with the vertical axis — the symmetry axis of the crossing-angle plane, of module boundaries and of the Roman-Pot cutout — fakes 0.05, four to eleven times the projected amplitudes (Figure 1d). The measurement therefore sorts the events of each reconstructed bin by spin state f , histograms $N_f(\phi')$, and forms the bin-wise ratio

$$R(\phi') = \frac{\sum_f w_f N_f(\phi')}{\sum_f N_f(\phi')}, \quad w_f = P_f - \bar{P}, \quad \bar{P} = \frac{\sum_f L_f P_f}{\sum_f L_f}, \quad R = \frac{\sigma_P^2 T(\phi')}{1 + \bar{P} T(\phi')}, \quad T = \kappa + \hat{A} \cos 2\phi', \quad \sigma_P^2 = \frac{\sum_f L_f (P_f - \bar{P})^2}{\sum_f L_f},$$

which is exact for yields $N_f = L_f \epsilon(\phi') \sigma [1 + P_f T]$ because $\sum_f L_f w_f = 0$: any efficiency common to the spin states cancels bin by bin, and a relative-luminosity error enters only through $\bar{P}$, as a ϕ' -independent constant plus a second-order rescaling. R is inverted for T bin by bin (with the exact Jacobian in the error propagation) and T fitted by weighted least squares to $\kappa + \hat{A} \cos 2\phi' + \hat{A}_s \sin 2\phi'$, the finite-bin dilution $\sin w/w$ divided out; the bin variances use the expected per-state counts so that a bin populated by one state alone keeps a finite weight. Two states of equal luminosity give $\delta \hat{A} = 2\sqrt{(2/N)/(P_+ - P_-)}$: the $m = \pm 1$ -rich state at $P_+ = +0.6$ alternated with the $m = 0$ -rich state at $P_- = -1.2$ is $1.5\times$ more precise than a single $P_{zz} = 0.6$ fill with the whole luminosity, and an unpolarized reference state would cost a factor two instead. κ is the transverse tensor rate shift of the b_1 sector, measured as a by-product. The cancellation holds only for an acceptance *common* to the states: a difference $\delta \epsilon_2$ of the $\cos 2\phi'$ harmonic between them enters as $\sum_f L_f (P_f - \bar{P}) \epsilon_f / \sigma_P^2 = \delta \epsilon_2 / (P_+ - P_-)$ at any luminosity split, so the states must alternate bunch by bunch within a fill, or the harmonic be stable to 10^{-4} between fills (§6). The same estimator in two azimuths serves the coherent channel (§4.5).

4.4 · Response, calibration and unfolding

The inclusive response is importance-sampled: every accepted cell of a 60×45 log-log grid receives 400 pseudo-events, log-uniform in the cell, with weight σ_c/n_c (n_c the draws inside the generator acceptance, so the cell rate is conserved; the amplitude is taken at the cell centre, worth $\leq 0.2\%$ at the sweet spots), and each passes through the chain of Schematic 1. For a reconstructed bin the response gives its purity (the fraction of its rate that originates in the true bin), its efficiency (the fraction of the true-bin rate reconstructed in it after ϵ_{eID}), and D , the rate- and ϵ_{eID} -weighted, ϕ' -diluted amplitude of the events landing in it over the amplitude of the true bin. The expected ϕ' counts per spin state are exact bin integrals of $\text{den}_f + P_f \hat{A} \cos 2\phi'$ over those events, with the ϕ' efficiency and the luminosity shares of the run plan; they are Poisson-fluctuated at 10 and 100 fb^{-1}/u and the fit is compared with the reconstructed-bin truth, which is what an unbiased estimator on this sample must return.

The amplitude is exactly linear in Δ — Δ enters the master formula in one place — so the response is a linear operator on Δ , $\text{fold}(\Delta) = \langle (\partial A / \partial \Delta) \Delta(x_{\text{true}}, Q^2_{\text{true}}) \rangle$ over the events of the bin, and the conversion $\Delta = \hat{A} K$ with $K = \Delta(x_c, Q^2_c) / \text{fold}(\Delta)$ is the bin-centering with migration included. Evaluating K with the shape that generated the pseudo-data is circular and reproduces the injected curve by construction; evaluating it with a wrong prior biases Δ by up to 20–24% over the plotted bins of a Q^2 slice, and by (−4.2, +8.0, −5.6, +4.9)% (model B) or (+8.6, +2.5, +7.3, +3.2)% (flat toy) at the four spots. Fitting a two-parameter tilt of the prior *through* the response per Q^2 slice on a bounded grid — the data choose the shape, the box guards the conditioning — reduces the residual to at most 5.7 / 5.7 / 5.8% (model B) and 4.3 / 4.3 / 3.6% (toy) over the plotted bins of the three slices, and to (−1.5, −1.8, −0.3, +0.5)% and (+4.6, +3.0, +1.8, −0.9)% at the spots. The Δ error bar then carries the statistics, the shape-fit error, the response Monte Carlo error (0.1–1.0% per plotted bin) and the spread over the priors the tilt cannot absorb, the last dominating in the best-measured bins.

4.5 • The coherent channel

The intact recoil keeps the beam rigidity ($A/Z = 2$, $x_p \leq 0.01$), so the pots resolve only its transverse displacement: the reconstruction returns the angle pair (θ_x, θ_y) at the interaction point relative to the beam axis — crossing angle and reference orbit drop out by construction — smeared by the bunch's own divergence, and $p_T = A p_u |\theta|$, $\varphi_t = \text{atan2}(\theta_y, \theta_x)$, $|\theta| = p_T^2$ with x_L set to one ($t_{\min} = M^2 x_p^2 / (1 - x_p) = 3 \times 10^{-3} \text{ GeV}^2$ at $x_p = 0.01$ is a known bias once x_p is measured from M_X). A recoil is tagged when it clears, per axis, the larger of the 10σ beam envelope and the pot aperture. At the Yellow Report optics with the measured aperture the tagged fraction is 7×10^{-8} , 6×10^{-27} and 2×10^{-17} at the three configurations (Figure 2c): the coherent channel exists only under the tagging optics of Report 1 §6.1, in which the horizontal β^* is raised to the optimum of acceptance $\times$ luminosity, the vertical plane stays at high acceptance, and the pots follow the envelope in both planes — a cutout of $0.33 \times 3.8 \text{ mrad}$ ($0.08 \times 0.93 \text{ GeV}$) at 5×40.8 with $\sigma_\theta = 33 / 380 \text{ } \mu\text{rad}$ (the horizontal including the beam's momentum spread through the pot dispersion), giving a 41% tag at 1/7.1 of the high-acceptance luminosity. The closure below is at that point.

With the recoil measured there are two physical azimuths besides the spin: $\alpha = \varphi_e - \varphi_S$ and $\beta = \varphi_t - \varphi_S$, and the spin-sorted yield is modelled as

$$N_f(\alpha, \beta) \propto \varepsilon(\alpha, \beta) [1 + u_1 \cos(\alpha - \beta) + u_2 \cos 2(\alpha - \beta) + P_f(\kappa + a_e \cos 2\alpha + a_t(t) \cos 2\beta + a_m \cos(\alpha + \beta))],$$

where a_t is the deformation term of the polarized-deuteron anchor [9] ($\propto |\theta|$ at low $|\theta|$, coefficient $2a_2$ in the anchor's convention, predicted to flip sign for ${}^6\text{Li}$), a_e the flat double-helicity-flip term that needs the lepton plane, a_m a nuisance, and u_1, u_2 the L–T and T–T' interference of unpolarized diffractive DIS [13,14]. The two mechanisms are separated by the azimuth they modulate and by their t dependence. The cutout is far from circular, so its acceptance carries a large $\langle \cos 2\beta \rangle$ of its own — −0.27 at the tagging cutout after the divergence smearing, −0.93 at the measured aperture — and no single-fill fit survives; the spin-state ratio of §4.3 in (α, β) cancels it bin by bin. Because the cutout correlates β with $|\theta|$ inside a reconstructed $|\theta|$ bin, the fit uses an acceptance-weighted basis from the response — the in-bin means of $\cos \beta$, $\sin \beta$, $\cos 2\beta$, $\sin 2\beta$ of the true β per reconstructed β bin, and the template $\langle (t/t_{\text{ref}}) \cos 2\beta \rangle$ of the assumed linear shape — so that the coefficients refer to the unsmeared truth and a_t is quoted at t_{ref} , the rate-weighted mean true $|\theta|$ of the bin; this is one coefficient per bin, the t dependence of a_t coming from the bins together. The three parity-forbidden partners $\sin 2\alpha$, $\sin 2\beta$ and $\sin(\alpha + \beta)$ are fitted alongside as a null test: a spin-axis error δ gives $\tan 2\delta$ in both azimuths, a pot roll δ_t gives $\tan 2\delta_t$ in β alone. α is integrated analytically per event; the coherent pseudo-experiment has no electron side, and x_p is sampled but never reconstructed (the M_X method would provide it).

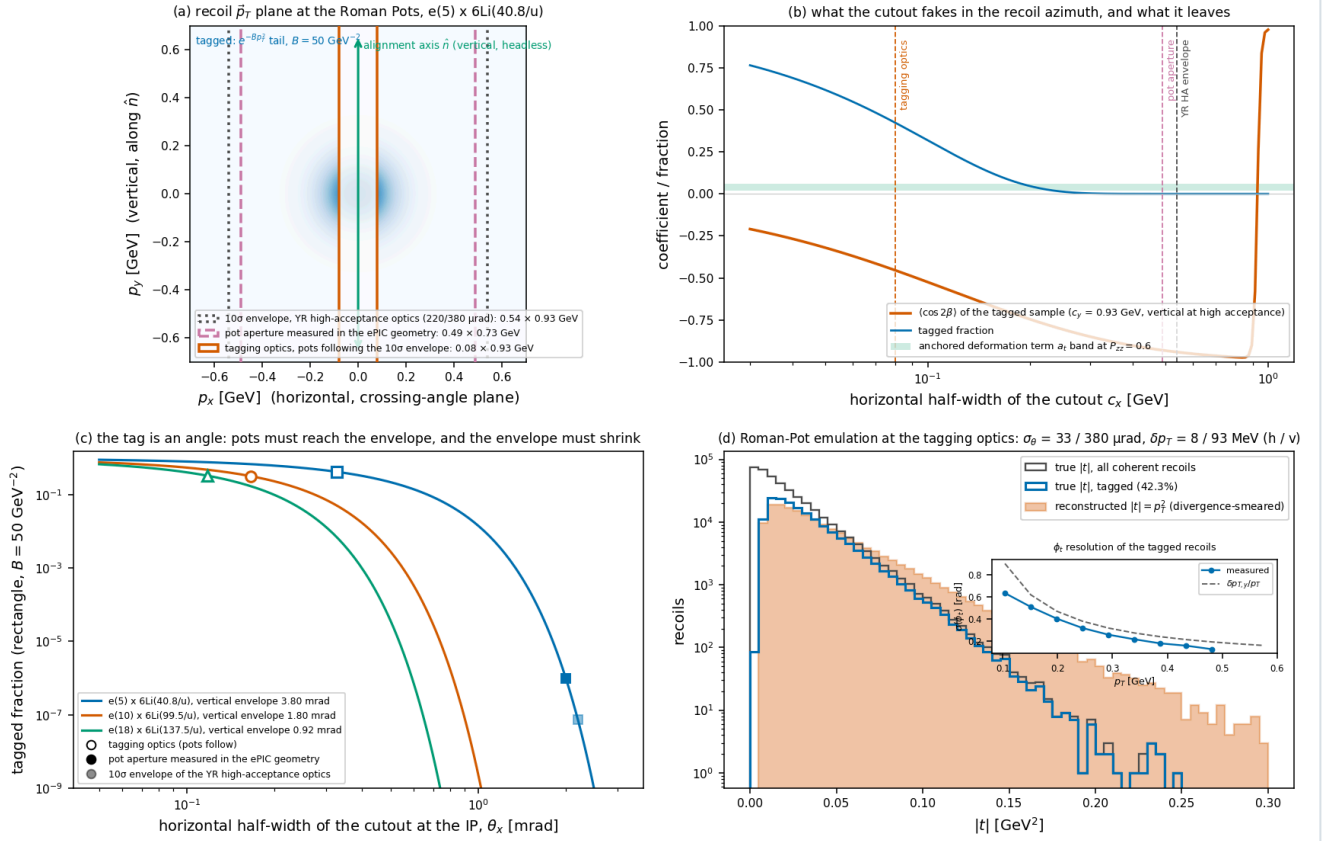


Figure 2 — the coherent measurement quantified (5×40.8 , `reco_chain_figures.py`). **(a)** The recoil p_T plane with the exponential coherent spectrum and the three cutouts: the Yellow Report high-acceptance 10σ envelope, the pot aperture measured in the ePIC geometry, and the tagging optics with pots following the envelope. **(b)** $\langle \cos 2\beta \rangle$ of the tagged sample about the vertical spin axis and the tagged fraction versus the horizontal half-width of the cutout, against the anchored deformation band. **(c)** Tagged fraction versus the horizontal half-angle at the interaction point for the three configurations, with the three cutouts marked. **(d)** The Roman-Pot emulation at the tagging optics: true, tagged and reconstructed $|t|$, and (inset) the ϕ_t resolution, dominated by the 93 MeV vertical divergence.

4.6 • What the analysis reads

The paths from counts to $\hat{A}$ and to a_t, a_e were traced assignment by assignment in the review (Appendix A). They consume: the spin state of each bunch crossing, the tensor polarization and luminosity share of each state (the run-plan bookkeeping, known to an experiment from the source pattern, the polarimeter and the luminosity monitor); the reconstructed electron four-vector; the hadronic sums and the calibration map at the reconstructed point; the Roman-Pot angle pair after the cut; and the simulation-derived corrections — purity, efficiency, K , the β basis, t_{ref} — in reconstructed bins. Generator truth is read to generate the pseudo-data, to build those corrections, and to plot the closure references; the analysis assumes the nuisances u_1, u_2 and the luminosity shares (as fit arguments, so that a wrong assumption can be exercised) and takes P_{zz} as exactly known (its scale is a normalisation, §6). The bins shown in the figures are selected on their expected count and statistical error alone.

5 · Closure at the reconstructed level

5.1 · Inclusive: money plots 5R and 7R

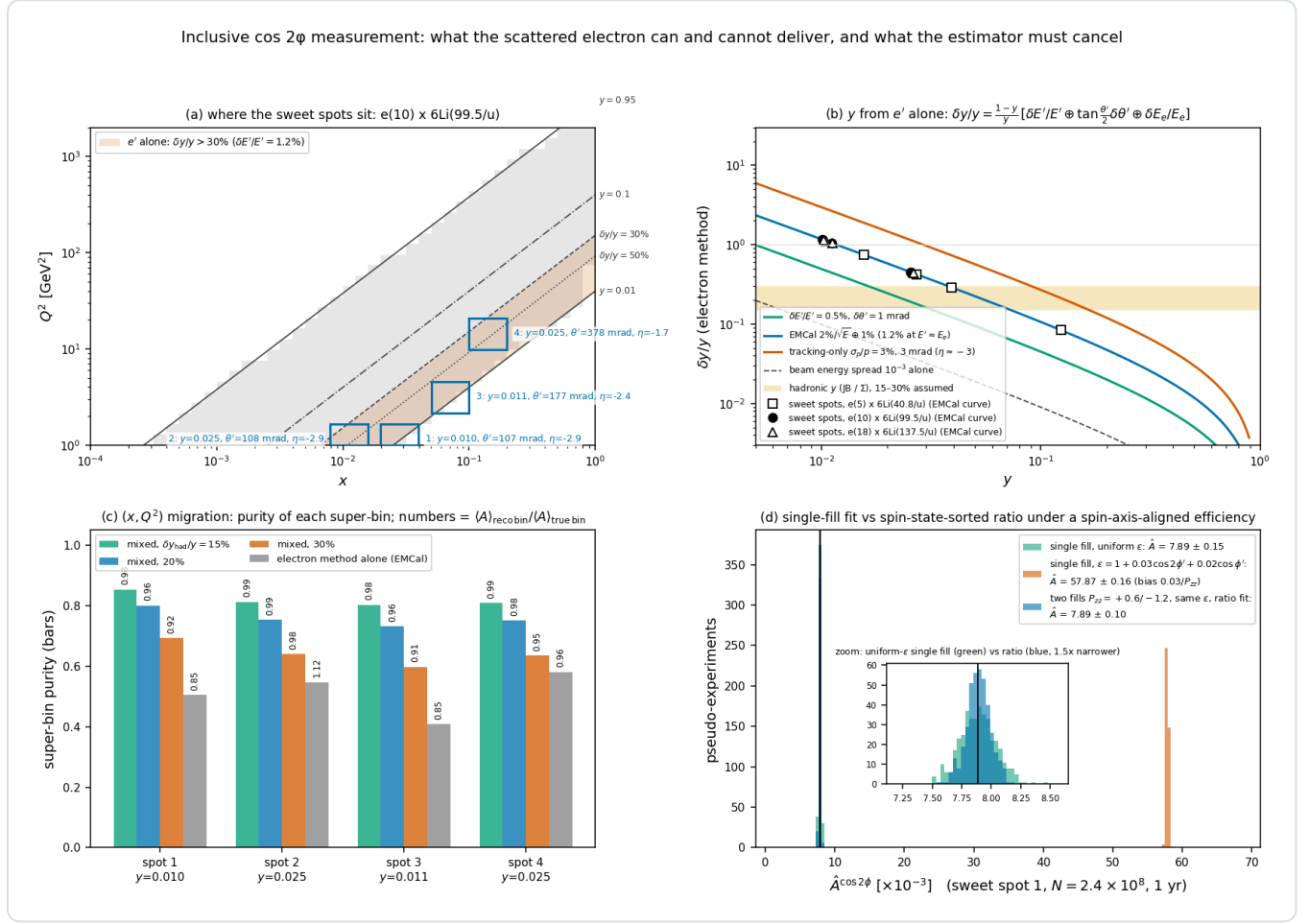
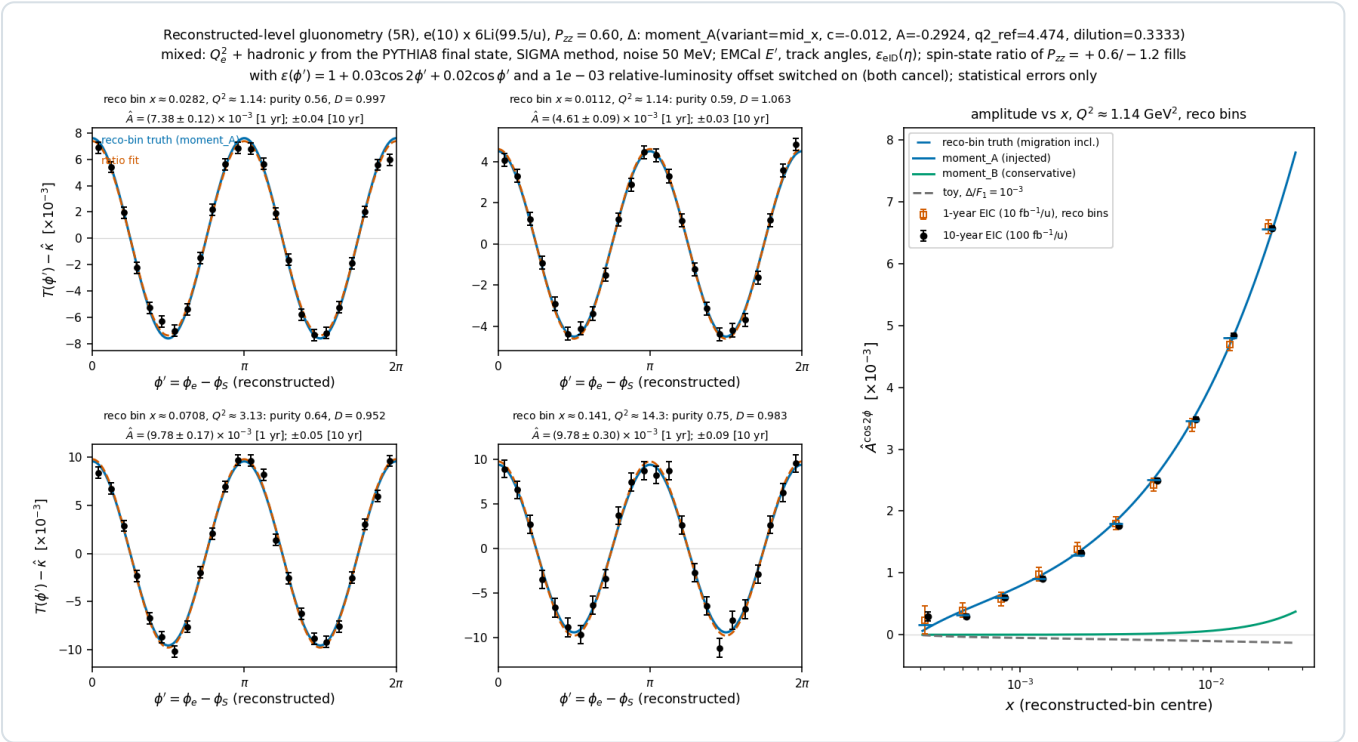


Figure 1 — the inclusive measurement quantified ($e10 \times {}^6\text{Li}99.5$, `reco_chain_figures.py`). **(a)** The analysis acceptance with iso- y lines and the four sweet-spot super-bins; the shaded band is where the electron alone gives $\delta y/y > 30\%$ at a 1.2% energy resolution. **(b)** The electron-method $\delta y/y$ versus y for calorimeter, tracking and beam-energy-spread inputs, with the sweet spots of the three configurations and the 15–30% hadronic band. **(c)** Purity of each super-bin for three Gaussian hadronic- y resolutions and for the electron method alone; the numbers are the reconstructed-bin over true-bin amplitude. **(d)** Four hundred pseudo-experiments of the spot-1 super-bin at one year: the single-fill fit under a uniform efficiency (green), the same under $\epsilon = 1 + 0.03 \cos 2\phi' + 0.02 \cos \phi'$ (vermillion, biased by $0.03/P_{zz}$), and the two-state ratio under that efficiency (blue, unbiased and $1.5\times$ narrower).

SWEET SPOT (RECO BIN)	25% GAUSSIAN HADRONIC-Y STAND-IN				PYTHIA 8 FINAL STATE, Σ METHOD, 50 MEV NOISE, CALIBRATED					$\Delta \hat{A}$ (10 YR) [10 ⁻⁵]	ΔA , REPORT 1 [10 ⁻⁴]
	PURITY	EFF.	D	$\hat{A} \pm \Delta \hat{A}$ (1 YR) [10 ⁻³]	N (1 YR)	PURITY	EFF.	D	$\hat{A} \pm \Delta \hat{A}$ VS RECO-BIN TRUTH [10 ⁻³]		
1: x = 0.028, Q ² = 1.14	0.66	0.42	0.92	6.88 ± 0.12 (7.00)	1.6×10 ⁸	0.56	0.31	1.00	7.38 ± 0.13 (7.61)	3.9	1.7
2: x = 0.011, Q ² = 1.14	0.63	0.59	0.99	4.30 ± 0.09 (4.20)	3.0×10 ⁸	0.59	0.56	1.06	4.61 ± 0.09 (4.50)	2.9	1.4
3: x = 0.071, Q ² = 3.13	0.69	0.38	0.90	8.71 ± 0.16 (9.07)	8.8×10 ⁷	0.64	0.32	0.95	9.78 ± 0.17 (9.56)	5.4	2.8
4: x = 0.141, Q ² = 14.3	0.69	0.66	0.96	9.07 ± 0.29 (9.23)	2.9×10 ⁷	0.75	0.71	0.98	9.78 ± 0.30 (9.41)	9.4	4.5

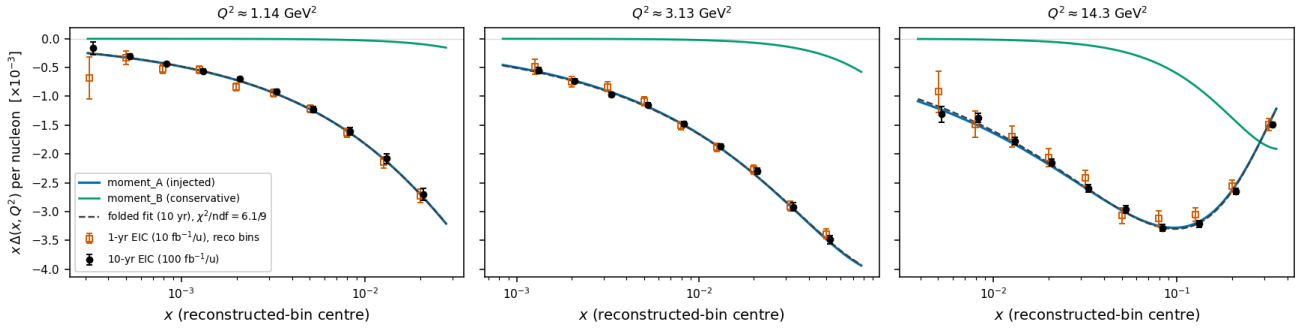
Table 4 — money plot 5R: the four sweet-spot super-bins in reconstructed (x , Q^2) at $e(10) \times {}^6\text{Li}(99.5)$, $P_{zz} = 0.6$, model A, mixed method, spin-state ratio of the $+0.6 / -1.2$ states with the acceptance harmonic and the 10^{-3} luminosity offset on, statistical errors only. The reconstructed-bin truth is in parentheses. With the PYTHIA final state uncalibrated the purities would be $0.42 / 0.53 / 0.49 / 0.68$ and $D = 0.79 / 0.84 / 0.83 / 0.95$; the errors are the same to within 7%. The last column is the single-state, truth-binned error of Report 1.

Three observations. The fit is unbiased: the eight pulls against the reconstructed-bin truth lie within 2 standard errors with the acceptance harmonic and the luminosity offset on, and the $\sin 2\phi'$ coefficients are consistent with zero. The errors are $0.59\text{--}0.69$ of the single-state truth-level values of Report 1, because the $1.5\times$ gain of the $m = 0$ -rich state outweighs the reconstruction efficiency of $0.3\text{--}0.7$ — presuming the source delivers that state at $|P_{zz}| = 1.2$ with the same purity as the ± 1 state at 0.6 . And the real final state costs purity rather than statistics: the $y \approx 0.01$ spots lose about 60% of their events to the reconstructed $y \geq 0.01$ cut and a further third of the remainder to the Σ resolution, which is why more of the answer passes through the bin-centering factor there. Unfolded to Δ (money plot 7R), the best reconstructed bins reach statistical errors $\delta\Delta = 2.3 \times 10^{-3}$ at $Q^2 = 1.14 \text{ GeV}^2$, 1.1×10^{-3} at 3.13 GeV^2 and 0.3×10^{-3} at 14.3 GeV^2 in one year ($1.7 / 1.7 / 6.9\%$ of Δ ; purities $0.44\text{--}0.56$; $2.5 / 1.2 / 0.3 \times 10^{-3}$ at $0.54\text{--}0.57$ with the stand-in). With the bin-centering factor fitted through the response the full one-year bars at those bins are $4.2 / 2.6 / 7.0\%$ and the ten-year ones $3.8 / 2.2 / 2.6\%$, the prior spread ($3.7 / 2.1 / 1.3\%$) dominating at the two lower Q^2 ; at $Q^2 = 1.14$ and 3.13 GeV^2 the ten-year data reject the flat-toy shape ($\chi^2/\text{ndf} = 31/9$ and $42/10$ against $6/9$ and $16/10$ for the injected one), at 14.3 GeV^2 no member of the family is distinguishable.



Money plot 5R — inclusive gluonometry at the reconstructed level ($e10 \times {}^6\text{Li}99.5$, $P_{zz} = 0.6$, model A, PYTHIA 8 final state with the Σ method and the calibrated hadronic scale, EMCal energy, track angles, $\epsilon_{\text{elD}}(\eta)$, spin-state ratio of the $+0.6 / -1.2$ states with the acceptance harmonic and luminosity offset on). Left: the acceptance-free modulation $T(\phi') - \hat{k}$ per unit P_{zz} with the reconstructed-bin truth (blue) and the fit (vermillion). Right: the amplitude in reconstructed x bins along the $Q^2 = 1.14 \text{ GeV}^2$ slice against the reconstructed-bin truth (blue ticks) and the model curves.

Reconstructed-level $\Delta(x, Q^2)$ extraction (7R), $e(10) \times 6\text{Li}(99.5/u)$, $P_{zz} = 0.60$: $\hat{\Delta} = \hat{A}K$, K from a 3-parameter $\Delta(x)$ shape fitted THROUGH the response (prior moment_A)
 · hadronic y from the PYTHIA8 final state, SIGMA method, noise 50 MeV; spin-state ratio estimator; dilution 1/3 incl.; stat. (+) shape-fit (+) response-MC (+) prior spread; area = S-S moment



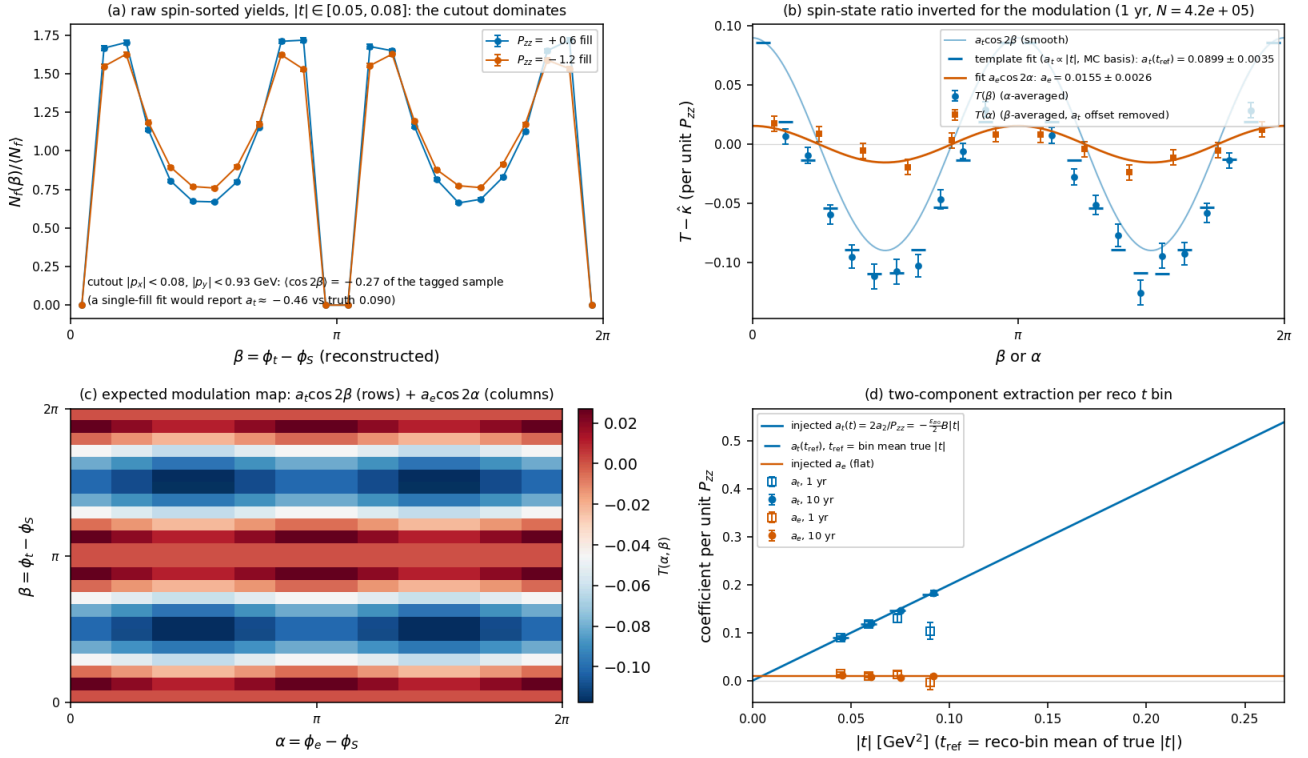
Money plot 7R — $\Delta(x, Q^2)$ from reconstructed bins. $x\Delta$ points at the three sweet-spot Q^2 slices from the same pseudo-measurements, converted with the bin-centering factor fitted through the response (§4.4), so that the error bars carry the statistics, the shape fit, the response Monte Carlo and the prior spread; one-year (open) and ten-year (filled) draws are independent.

5.2 · Coherent: money plot 6R

CONFIGURATION	RECO T BIN [GeV²]	$N_{\text{TAG}}, 1$ YR	$A_T(T_{\text{REF}})$ INJECTED	FIT, 1 YR	ENSEMBLE MEAN (20 EXP.)	FIT, 10 YR	A_E FIT, 1 YR (INJ. 0.010)	ENSEMBLE MEAN	A_E FIT, 10 YR
5×40.8 , tag 41%, $N_{\text{tag}} = 2.5 \times 10^6/\text{yr}$	0.05– 0.08	4.2×10^5	0.090	0.0899 ± 0.0035	0.0899 (σ 0.0032)	0.0892 ± 0.0011	0.0155 ± 0.0026	0.0092 (σ 0.0033)	0.0107 ± 0.0008
	0.08– 0.12	1.8×10^5	0.118	0.1184 ± 0.0048	0.1170 (σ 0.0038)	0.1202 ± 0.0015	0.0103 ± 0.0040	0.0097 (σ 0.0037)	0.0084 ± 0.0012
	0.12– 0.17	5.4×10^4	0.147	0.131 ± 0.009	0.139 (σ 0.008)	0.1465 ± 0.0027	0.012 ± 0.007	0.0101 (σ 0.0069)	0.0066 ± 0.0022
	0.17– 0.25	1.3×10^4	0.180	0.104 ± 0.018	0.114 (σ 0.023)	0.183 ± 0.006	−0.004 ± 0.014	0.008 (σ 0.012)	0.0095 ± 0.0044
18×137.5 , tag 33%, $N_{\text{tag}} =$ $6.0 \times 10^6/\text{yr}$	0.05– 0.08	1.1×10^6	0.100	0.1005 ± 0.0023	0.0998 (σ 0.0023)	0.0992 ± 0.0007	0.0132 ± 0.0016	0.0096 (σ 0.0019)	0.0104 ± 0.0005
	0.08– 0.12	4.1×10^5	0.137	0.1333 ± 0.0033	0.1368 (σ 0.0041)	0.1378 ± 0.0011	0.0148 ± 0.0026	0.0091 (σ 0.0028)	0.0108 ± 0.0008
	0.12– 0.17	1.0×10^5	0.179	0.1766 ± 0.0064	0.1776 (σ 0.0057)	0.1789 ± 0.0020	0.0137 ± 0.0052	0.0097 (σ 0.0027)	0.0095 ± 0.0016
	0.17– 0.25	1.9×10^4	0.228	0.221 ± 0.015	0.207 (σ 0.016)	0.226 ± 0.005	0.008 ± 0.012	0.0103 (σ 0.0102)	0.0130 ± 0.0037

Table 5 — money plot 6R: the two-azimuth template fit per reconstructed |t| bin at the tagging optics (coefficients per unit P_{zz} ; $\epsilon_{B0} = -0.08$, $B = 50 \text{ GeV}^{-2}$; $u_1 = 0.05$, $u_2 = 0.02$; luminosity at the tagging optics, 1/7.1 and 1/9.5 of the high-acceptance value; 6×10^6 response recoils; `money_cos2phi_coherent_reco.py --optics tagging`). The "fit" columns are one pseudo-experiment; the ensemble columns are the mean and spread of twenty one-year pseudo-experiments on the same response and say whether the estimator is biased. With more than 10^5 tagged recoils per bin it is not: the means agree with the injected values within their standard errors for a_t and a_e alike, and the single-experiment a_e of the first bin, 1.7 standard errors high, is the fluctuation the ensemble shows it to be. Below that count the bin-wise ratio of Poisson counts is biased low (−5% and −37% in the two upper bins of 5×40.8 at one year), and the same bins close at ten years; 6×12 instead of 12×24 (α , β) bins reduce the two biases to −4% and −10%, and a likelihood fit would remove them (§7). The sine columns are consistent with zero throughout and resolve a pot roll to 16–24 mrad per year in the three lower bins.

Coherent $e^6\text{Li} \rightarrow e^7\text{Li}^6\text{Li}(\text{g.s.})$ at the reconstructed level (6R), $e(5) \times 6\text{Li}(40.8/\text{u})$, $P_{zz} = 0.60$: $\sigma_\theta = 33 / 380 \mu\text{rad}$ (h / v), rectangle cutout $|p_x| < 0.08$, $|p_y| < 0.93 \text{ GeV}$ [tagging optics]
 $N_{\text{tag}} = 2.5\text{e}+06$ (1 yr) vs $5.8\text{e}+06$ with the constant 0.20 GeV cut at L_{HA} ; deformation $c_2 = 2a_2$ (Eq. 9 convention), $a_\theta = 0.010$
 $u_1 = 0.05$, $u_2 = 0.02$ (ZEUS LPS 1σ bounds); two-fill ratio fit with the MC template basis; statistical errors only



Money plot 6R — the coherent channel at the reconstructed level (5×40.8 at the tagging optics). **(a)** Raw spin-sorted yields versus the reconstructed recoil azimuth β in the lowest $|t|$ bin: the cutout shape dominates both states identically. **(b)** The modulations $T(\beta)$ and $T(\alpha)$ from the spin-state ratio, the template-fit prediction per β bin (ticks) and the smooth $a_t \cos 2\beta$ and $a_e \cos 2\alpha$ curves. **(c)** The expected modulation map in (α, β) . **(d)** $a_t(t_{\text{ref}})$ and a_e per reconstructed $|t|$ bin (1 yr open, 10 yr filled) against the injected curves.

6 • Systematic effects and the specifications they imply

Each effect was measured on the chain by holding the response fixed and varying one input with common random numbers (inclusive: `money_cos2phi_reco.py --syst-scan`; coherent: exact expected counts with the assumption changed on the fit side only). The response's own Monte-Carlo floor is 0.06–0.28% on Δ at the four spots.

EFFECT	SIZE	ON THE RESULT
electron energy scale $\pm 1\%$	unknown to the analysis	Δ by 0.3–1.0% at the four spots ($d \ln x / d \ln E' = 2 - y$ with the Σ method)
hadronic energy scale, residual after calibration	+2%	$\hat{A}$ by +0.2 to +1.2% (0.1–0.6% per 1%)
hadronic resolution mis-modelled	generated at 30%, corrected with 25%	Δ by 0.5–1.4%
η -dependent calorimetry; ϵ_{elD} η tilt of 0.05	—	$\leq 0.1\%$; 0.02%
bin-centering model dependence	a wrong prior	4–9% at the spots with an assumed K , $\leq 1.8\%$ (model B) / $\leq 4.6\%$ (toy) with the shape fitted through the response
acceptance harmonic differing between the spin states	$\delta\epsilon_2 = 10^{-3}$	fakes $\hat{A} = 5.6 \times 10^{-4}$ — 6–13% of the projected amplitudes, 2–6 one-year errors

relative luminosity between the states	10^{-3} on the ratio	0.03% (1/3 per unit ratio error, a constant in R)
P_{zz} scale	$\delta P_{zz}/P_{zz}$	1:1 on Δ , exactly (the ratio carries one power of the assumed scale against σ_P 's two); a pure normalisation — every shape claim is immune
coherent: cutout differing between the states	horizontal half-width differing by 10^{-3} (10^{-2}) between the states	a_t by $-0.8 / -0.5 / -0.1 / -0.3\%$ ($-7.6 / -4.5 / -3.6 / -2.9\%$) in the four $ t $ bins, a_e untouched — at the tagging cutout, whose horizontal edge sits inside a shallow part of the spectrum, a $\sim 10^{-3}$ stability of the envelope is a per-mille effect (it was +19% under the earlier slot geometry)
coherent: u_2 assumed wrong	$\delta u_2 = 0.024$ (a ZEUS 1σ)	a_e by $+0.0007$ to $+0.0009$ (7-9%, 0.3 one-year errors), a_t untouched; a wrong u_1 moves only the nuisance a_m — first order through $a_t \delta u_2 (\cos 2\beta)$, to be measured in situ from the spin-averaged counts
coherent: relative luminosity	10^{-3}	0.1% on a_t , nothing on a_e

Table 6 — the systematic effects measured on the chain. Not modelled, and outside every error bar here: radiative corrections (uncalculated for tensor observables), backgrounds, and the F_2/R model (§7).

Two of the resulting specifications are this programme's to set, because no EIC document addresses a tensor-polarized beam: the tensor-polarization scale, $\delta P_{zz}/P_{zz} \leq 5\%$ (3% has been reached on stored deuterons at the Nuclotron and on the HERMES storage cell, 8–10% is where solid-target work sits; Report 1 §7), propagating 1:1 into the magnitude of Δ and not at all into its shape; and the relative luminosity between the tensor states, $\leq 10^{-3}$, which costs 0.03% and disappears with bunch-by-bunch alternation. The third is the acceptance-harmonic stability of 10^{-4} between the states, or the alternation that makes it exact. The others are asks with owners: $P_{zz} \geq 0.8$ from the source; a hadronic-method $\delta y/y \leq 0.3$ at $y \approx 0.01$ from ePIC full simulation, which turns on the calorimeter noise floor at $\Sigma_h = 0.2$ –0.5 GeV; the lithium tagging optics and pots that follow it from C-AD and the far-forward group; charge identification in the pots at the 10^{-3} level for the $\alpha \leftrightarrow {}^6\text{Li}$ separation; and a far-forward reconstruction that knows what lithium is, which is what this development supplies.

7 • What the chain does not model

Radiative corrections are absent: the Σ -method y and the mixed x are insensitive to collinear initial-state radiation, the Q^2 label is not, and tensor radiative corrections have not been calculated (plans/07 WP4). Backgrounds are absent: no electron-identification purity or pair-symmetric background inclusively, no $\alpha + d$ breakup, vetoes or Z-identification in the coherent sample, and no beam-gas or pile-up. The coherent pseudo-experiment has no electron side and no x_p or Q^2 binning; its template assumes the linear t shape it fits; and its bin-wise ratio estimator is biased below ~ 30 counts per (α, β) bin, which a coarser binning or a likelihood fit would remove. The structure functions are placeholders behind grid and R1998 backends. The track angular resolution and the calorimeter noise floor are stand-ins that the ePIC full simulation would replace, and the pot aperture was measured in a geometry that has since moved. The tagging optics is a proposal whose transport has not been computed. Each of these is listed in Table 2 or §6 with its leverage, and none changes the closure of §5, which is a statement about the chain given its inputs.

References

- [1] R. L. Jaffe, A. Manohar, *Nuclear gluonometry*, Phys. Lett. B 223 (1989) 218.
- [2] P. Hoodbhoy, R. L. Jaffe, A. Manohar, *Novel effects in deep inelastic scattering from spin-one hadrons*, Nucl. Phys. B 312 (1989) 571.
- [3] A. Bacchetta, M. Diehl, K. Goeke, A. Metz, P. J. Mulders, M. Schlegel, *Semi-inclusive deep inelastic scattering at small transverse momentum*, JHEP 02 (2007) 093, hep-ph/0611265 (Eqs. 2.4–2.5).
- [4] A. Bacchetta, U. D'Alesio, M. Diehl, C. A. Miller, *Single-spin asymmetries: the Trento conventions*, Phys. Rev. D 70 (2004) 117504; M. Diehl, S. Sapeta, Eur. Phys. J. C 41 (2005) 515.
- [5] F. Jacquet, A. Blondel, in *Proceedings of the Study of an ep Facility for Europe*, DESY 79/48 (1979) 391.
- [6] U. Bassler, G. Bernardi, *On the kinematic reconstruction of deep inelastic scattering at HERA: the Σ method*, Nucl. Instrum. Meth. A 361 (1995) 197.

- [7] R. Abdul Khalek et al., *Science requirements and detector concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447.
- [8] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (far-forward acceptance model).
- [9] H. Mäntysaari, F. Salazar, B. Schenke, C. Shen, W. Zhao, *Spatial imaging of polarized deuterons at the Electron-Ion Collider*, Phys. Lett. B 858 (2024) 139053, arXiv:2408.13213.
- [10] W. Chang, E.-C. Aschenauer, A. Jentsch, A. Kumar, Z. Tu, Z. Yin, Phys. Rev. D 113 (2026) 032018, arXiv:2511.05638 (IR-8 secondary focus).
- [11] This repository: Report 1 ([reports/cos2phi_money_plots_report](#)), Report 3 ([reports/eic_epic_reference](#)), [evgen/polligen/{reco,hfs,recopseudo}.py](#) , [docs/reproduction_manual.md](#) , [docs/code_review_2026-08-28.md](#) .
- [12] W. Cosyn, B. Roldan Tomei, A. Sosa, A. Zec, *Polarization options in inclusive DIS off tensor polarized deuteron*, Eur. Phys. J. A 61 (2025) 83, arXiv:2410.12764.
- [13] ZEUS Collaboration, S. Chekanov et al., Nucl. Phys. B 816 (2009) 1, arXiv:0812.2003 (leading-proton azimuthal asymmetries; beam-spread-dominated LPS resolution).
- [14] N. N. Nikolaev, A. V. Pronyaev, B. G. Zakharov, arXiv:hep-ph/9812212 (the y dependence of the diffractive azimuthal terms).
- [17] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (beam effects dominate the far-forward momentum smearing).
- [18] ATHENA Collaboration, JINST 17 (2022) P10019, arXiv:2210.09048 (Sec. 3.1, Fig. 22: hadronic-method resolution at low y ; Table 5: electron identification).
- [19] S. Maple (ePIC), *Tracking and inclusive DIS reconstruction with the ePIC detector at the EIC*, seminar, University of Birmingham, 11 December 2024 (the 25% hadronic smearing of the kinematic-fit study).
- [20] M. Arratia, D. Britzger, O. Long, B. Nachman, Nucl. Instrum. Meth. A 1025 (2022) 166164, arXiv:2110.05505 (Table 1: the classical methods; Sec. 5: noise and acceptance at low y).
- [21] K. Abe et al. (E143), Phys. Lett. B 452 (1999) 194, arXiv:hep-ex/9808028 (the R1998 world fit).
- [22] T. Sjöstrand et al., *An introduction to PYTHIA 8.2*, Comput. Phys. Commun. 191 (2015) 159; PYTHIA 8.3 (2022), the version used is 8.311.

Appendix A · Revision history and the 2026-08-28 review

DATE	REVISION
2026-08-28	Rewritten after a full review of the toolchain (nine independent readings of the code against the questions: is each step right, does any analysis step read information an experiment cannot have, is every assumption stated; findings adversarially verified and recorded in docs/code_review_2026-08-28.md). What was wrong and is fixed: the hadronic-scale calibration was looked up at each pseudo-event's true (x, Q^2) cell — now a map in reconstructed $(x_{\text{mixed}}, Q_e^2)$ bins built from the library's own reconstructed events, purities 0.56 / 0.59 / 0.64 / 0.75 against 0.52 / 0.59 / 0.59 / 0.76 before; the PYTHIA library carried PYTHIA's default $\hat{m} \geq 4$ GeV, which removed $x < 16/s$ (39% of the selected rate) — regenerated with $\hat{m} \geq 0.5$ GeV; the estimator's bin variance vanished for a bin populated by one spin state (an infinite weight, and the "rank-deficient" t bins of the earlier Table 3) — now the expected-count form; the ratio inversion's error Jacobian dropped a second-order term; the hadron acceptance was applied in the head-on frame (+0.7–1.5% of Σ_h in the lab); the library's target-mass term was not carried onto the pseudo-events (2% of Σ_h at $y = 0.01$); the $e + p / e + n$ merge was by count rather than cross section; the beam-energy spread had the same sign on Q_e^2 and $1 - y_e$; the bins of the 5R panels were selected partly on the truth amplitude; the coherent pseudo-experiment ran on the legacy 73 μrad divergence with a 0.73 mrad "near-beam approach" — now the tagging optics of Report 1 with the Yellow Report divergences, at all three configurations, with a per-fill perturbation that acts on the binding cut; and the "residual template bias" on a_e was a 2σ fluctuation, shown by the ensemble of Table 5. The stand-in numbers of Table 4 did not move; the calibrated ones moved within their errors. Stale statements removed: the pre-correction sweet spots of the old Table 1, coherent numbers at 50 GeV/u, the 73/149 μrad footer, the 25% stand-in as the published hadronic y, the $x \rightarrow x/(1 - z)$ statement about radiation.
2026-08-24 – 27	First version and its revisions: the covariant azimuth and the head-on frame, the electron-method limit and the hadronic methods, the spin-state ratio, the coherent two-azimuth fit with the acceptance-weighted template basis and the sine null columns, the PYTHIA 8 final state through the hadron-side response, the Roman-Pot aperture measured in the ePIC geometry, the forward-folded bin-centering fit, the per-fill acceptance and the assumed nuisances as fit arguments, the y-matched beam energies, the 1:1 propagation of the P_{zz} scale, and the consolidated assumptions and specifications. The numbers of those revisions that this one supersedes are recorded in plans/00 and plans/08.

Conventions: head-on frame with the ion along $+z$ and the electron along $-z$; θ' is the scattered-electron angle from the electron beam; per-nucleon (x, y, s) throughout; whole-nucleus four-momenta with the physical nuclear mass where a mass matters. Beam divergences are the Yellow Report Table 10.1 proton values with the y -matched species step, and the tagging optics of Report 1 §6.1 where stated. Built by [reports/build_report.py](#); figures from the [evgen/scripts](#) named in the text; display mathematics typeset at build time.